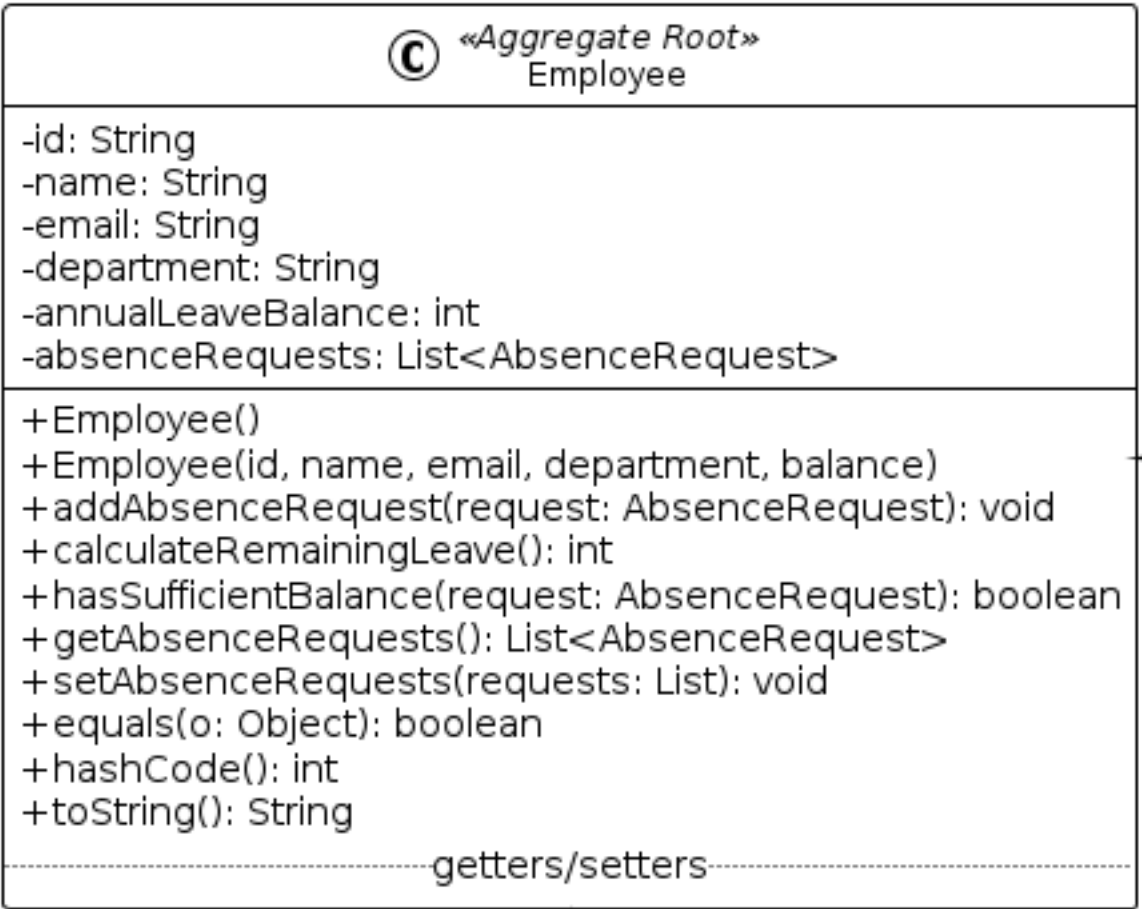
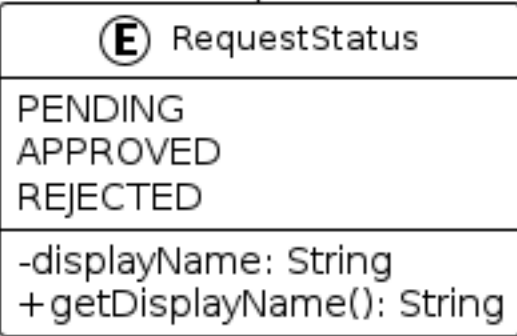
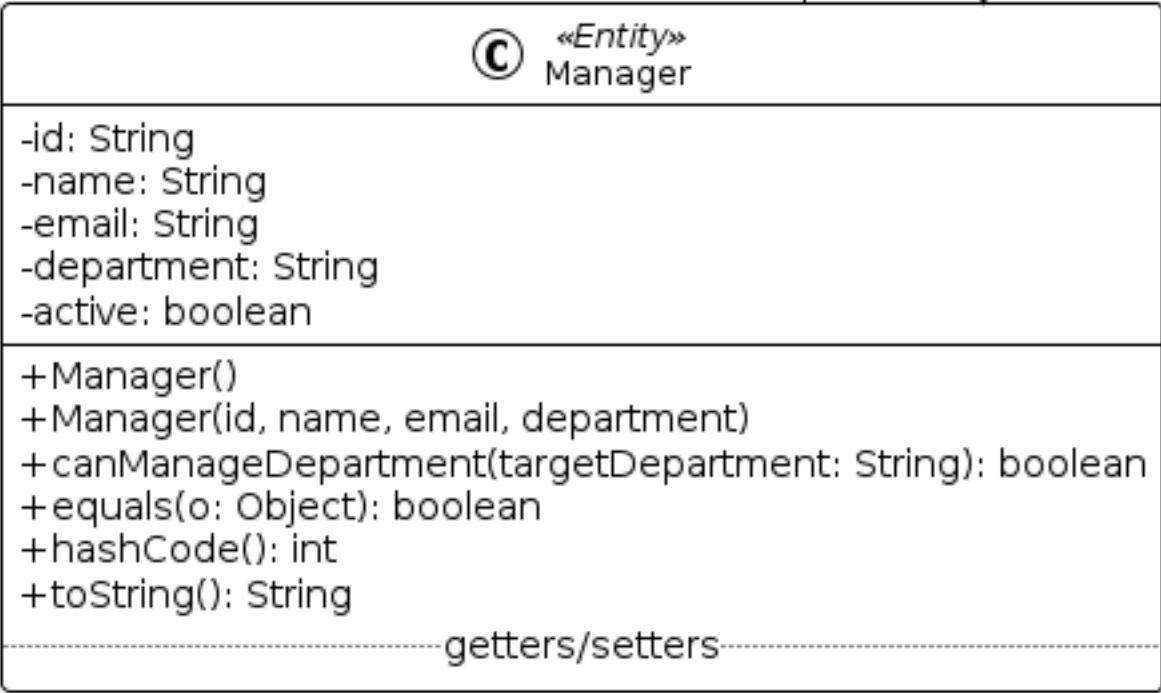
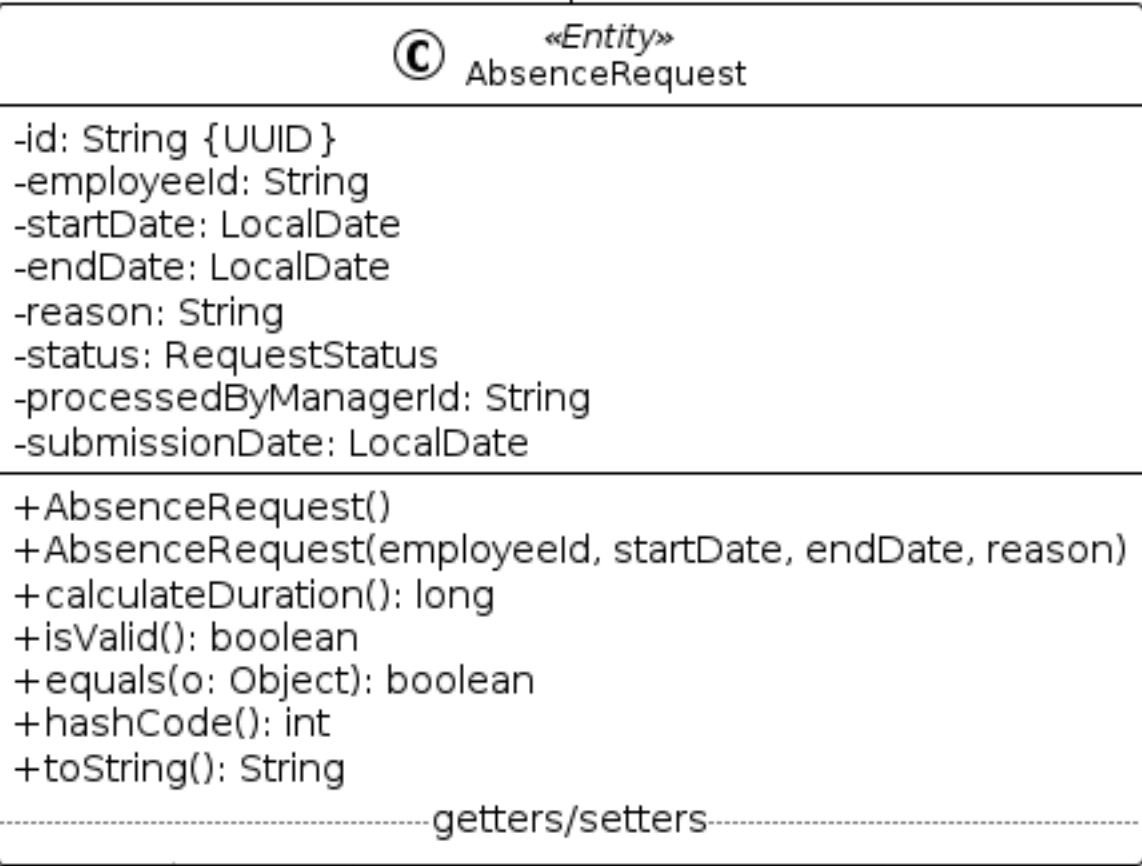


Domain Model - Entity Layer (com.absence.domain.model)

Domain Model



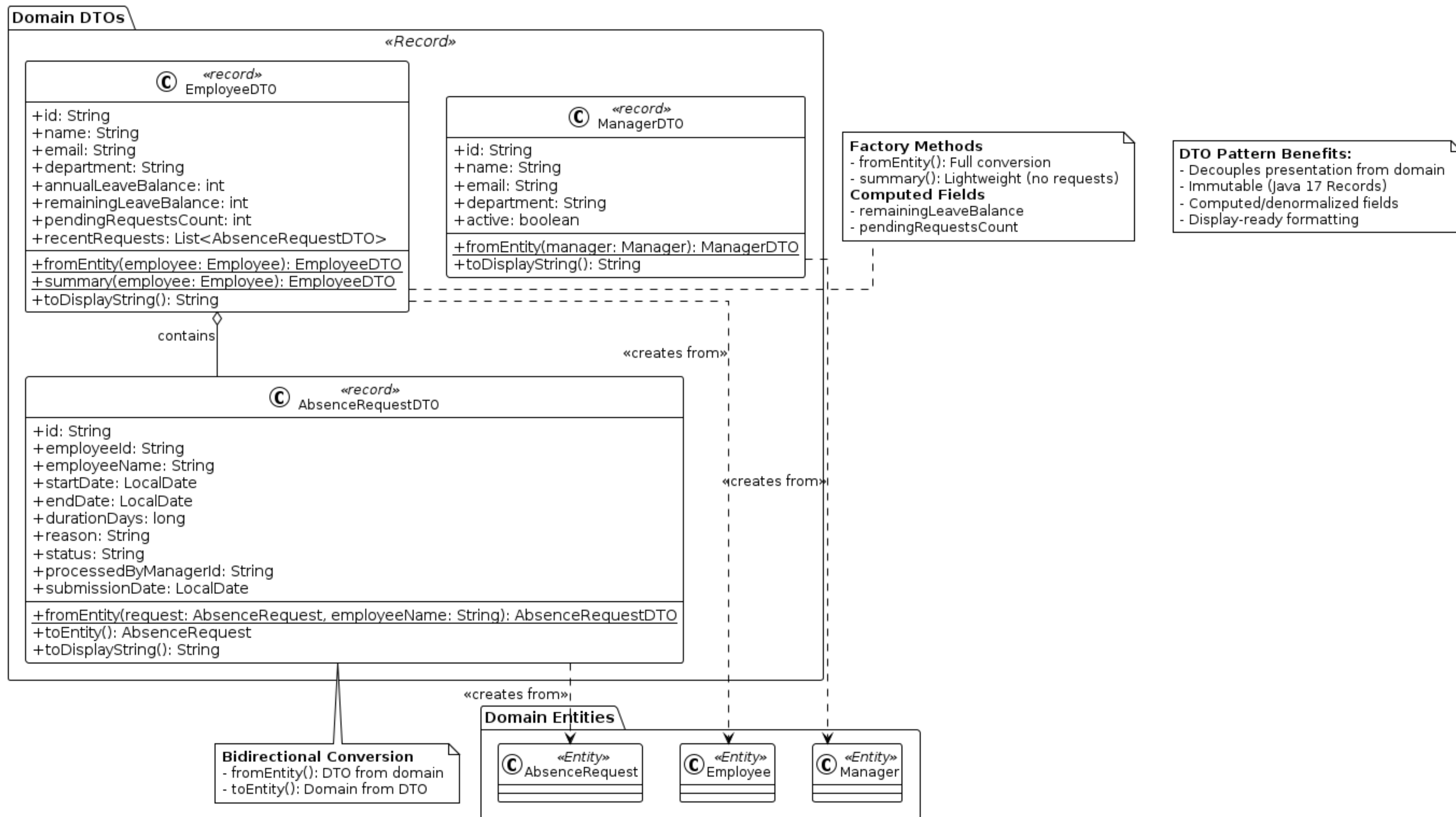
Aggregate Root
Manages collection of AbsenceRequest entities
Business logic for leave balance calculation



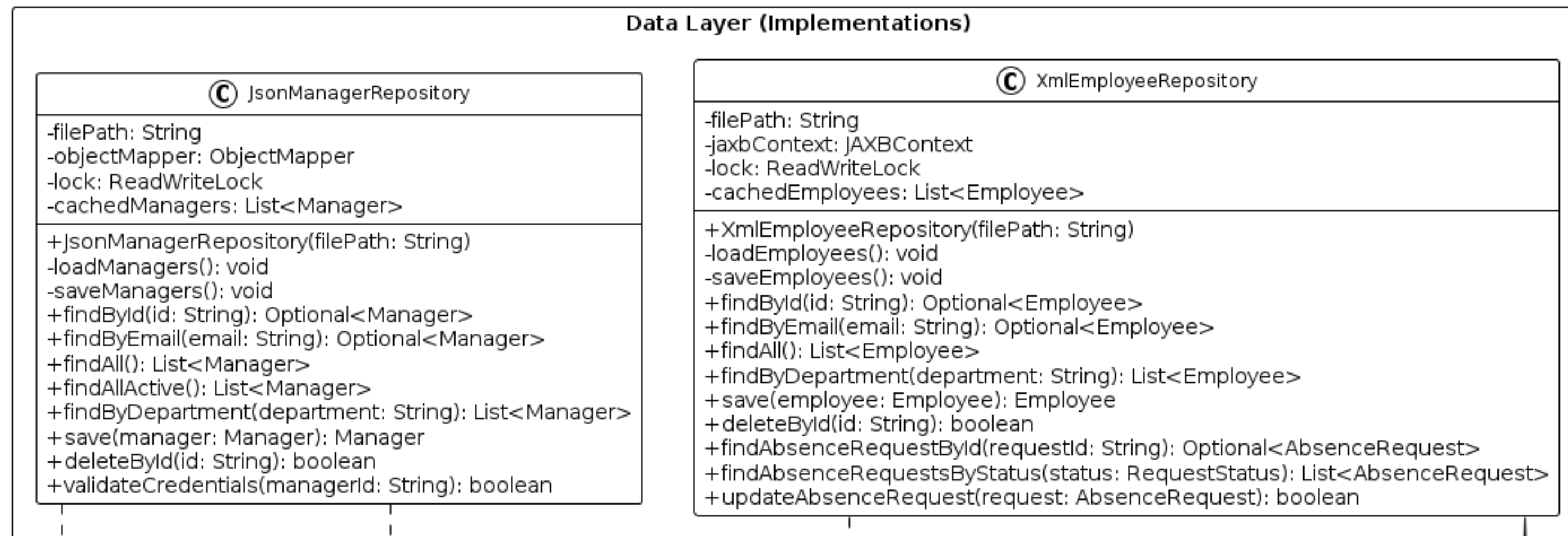
Business Rules:
- endDate >= startDate
- startDate >= today
- Auto-generates UUID
- Auto-sets submissionDate

Authorization Logic:
canManageDepartment returns true if department is null (manages all) or matches

Data Transfer Objects (com.absence.domain.dto)



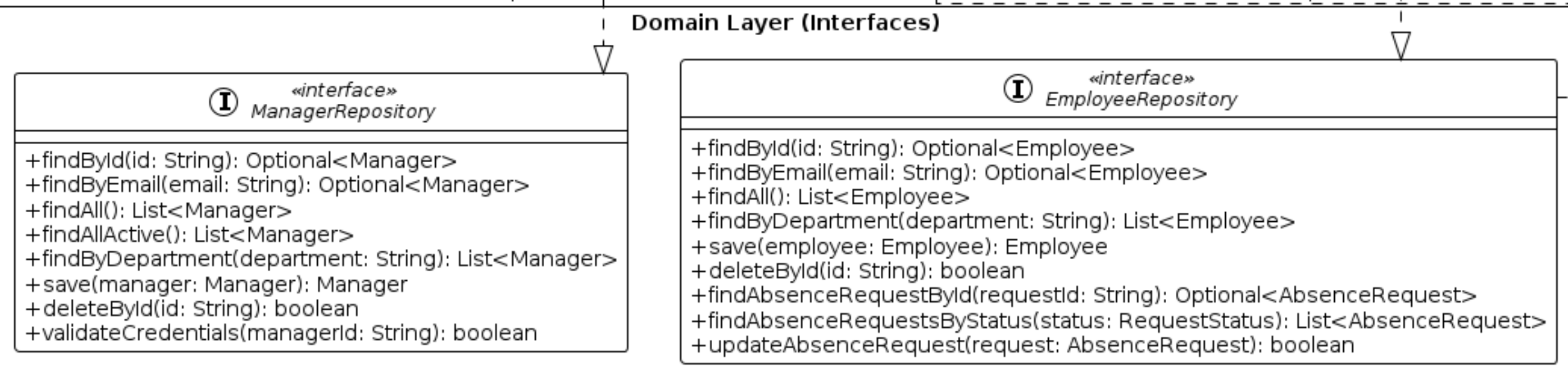
Repository Pattern - Data Access Layer



implements

implements

Domain Layer (Interfaces)

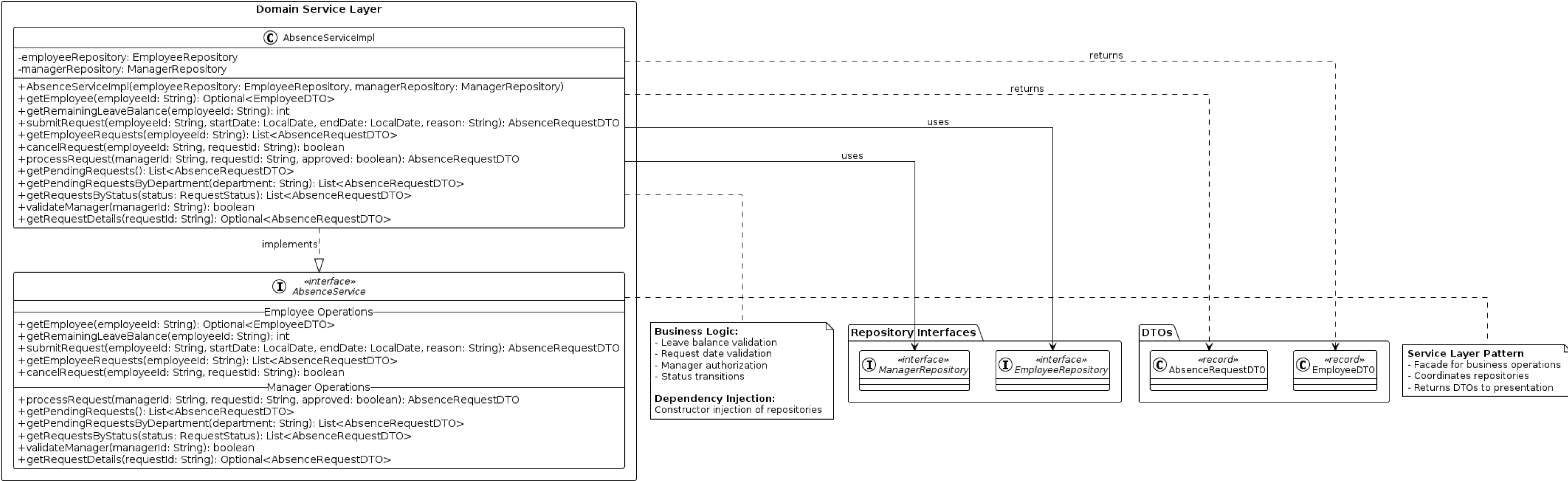


Thread Safety:
ReadWriteLock for concurrent access
Caching:
In-memory cache for performance
Persistence:
Jackson JSON serialization

Thread Safety:
ReadWriteLock for concurrent access
Caching:
In-memory cache for performance
Persistence:
JAXB XML serialization

Repository Pattern
Domain layer defines contract
Data layer provides implementation

Service Layer - Business Logic



Presentation Layer - Controllers

CLI Presentation

Ⓒ ManagerCommandLineInterface

-absenceService: AbsenceService
-reader: BufferedReader
-output: PrintStream
-currentManagerId: String
-running: boolean

+ManagerCommandLineInterface(absenceService: AbsenceService)
+ManagerCommandLineInterface(absenceService: AbsenceService, reader: BufferedReader, output: PrintStream)
+start(): void
-printWelcome(): void
-authenticate(): boolean
-printMenu(): void
-processChoice(choice: String): void
-viewPendingRequests(): void
-viewRequestsByStatus(): void
-approveRequest(): void
-rejectRequest(): void
-viewRequestDetails(): void
-exit(): void
-displayRequestList(requests: List<AbsenceRequestDTO>): void
-truncate(str: String, maxLength: int): String

Web Presentation

Ⓒ «@WebServlet» EmployeeWebController

-absenceService: AbsenceService

+EmployeeWebController()
+EmployeeWebController(absenceService: AbsenceService)
+init(): void
+setAbsenceService(absenceService: AbsenceService): void
#doGet(request: HttpServletRequest, response: HttpServletResponse): void
#doPost(request: HttpServletRequest, response: HttpServletResponse): void
#doDelete(request: HttpServletRequest, response: HttpServletResponse): void
-handleGetEmployee(employeeId: String, response: HttpServletResponse): void
-handleGetRequests(employeeId: String, response: HttpServletResponse): void
-handleSubmitRequest(employeeId: String, request: HttpServletRequest, response: HttpServletResponse): void
-handleCancelRequest(employeeId: String, requestId: String, response: HttpServletResponse): void
-sendJsonResponse(response: HttpServletResponse, status: int, json: String): void
-sendError(response: HttpServletResponse, status: int, message: String): void
-formatEmployeeJson(employee: EmployeeDTO): String
-formatRequestJson(request: AbsenceRequestDTO): String
-formatRequestListJson(requests: List<AbsenceRequestDTO>): String
-escapeJson(value: String): String

CLI Controller (MVC)

- Manager-facing interface
- Menu-driven interaction
- Text table formatting

Operations:

- View pending requests
- Approve/Reject requests
- View request details

REST Controller (MVC)

URL: /absence/employee/*

Endpoints:

GET /{id} - Employee info
GET /{id}/requests - List requests
POST /{id}/request - Submit request
DELETE /{id}/request/{rid} - Cancel

Response Format:

JSON with manual serialization

uses

Service Layer

Ⓘ «interface» AbsenceService

displays

uses

returns JSON!

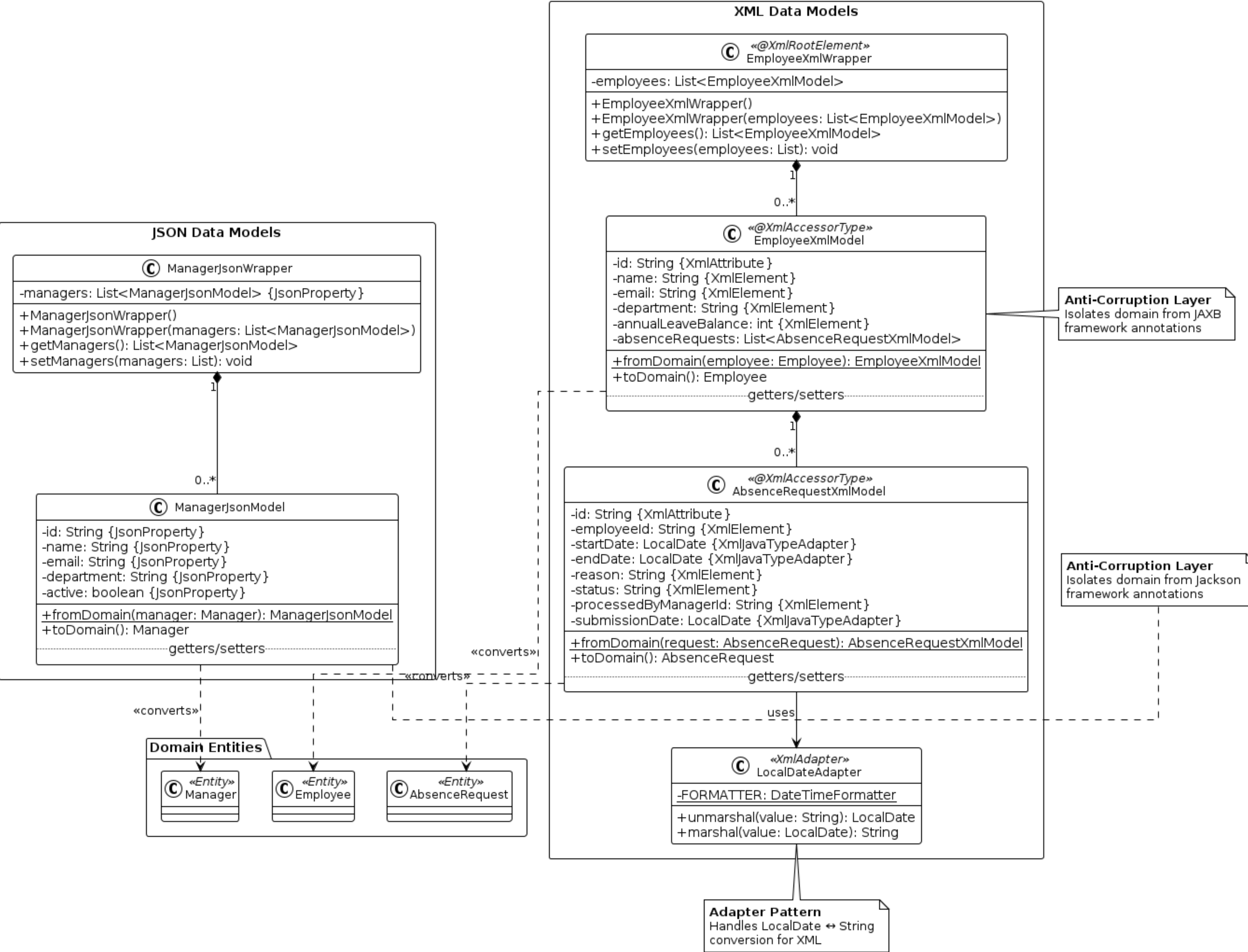
returns JSON!

DTOs

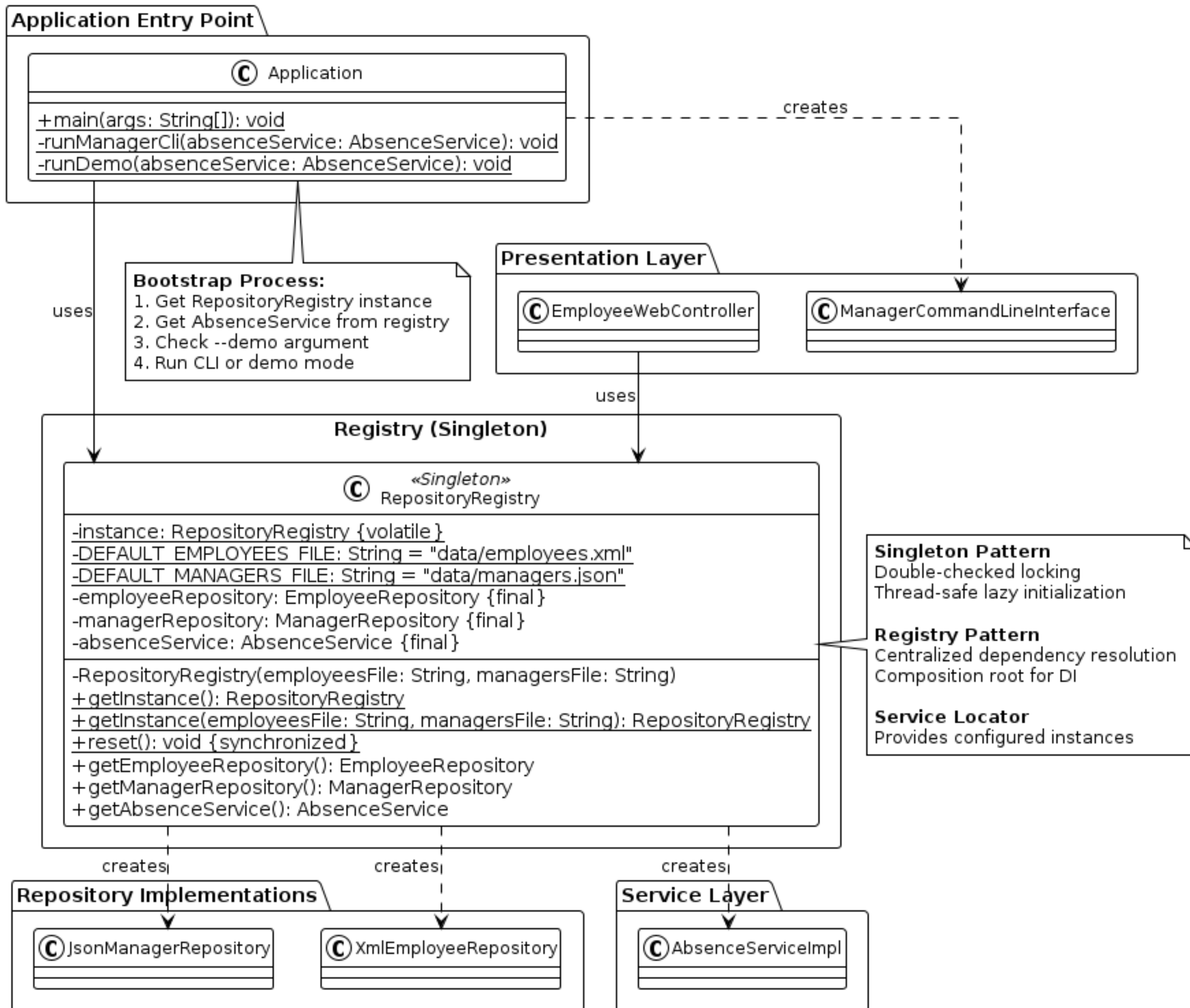
Ⓒ «record» AbsenceRequestDTO

Ⓒ «record» EmployeeDTO

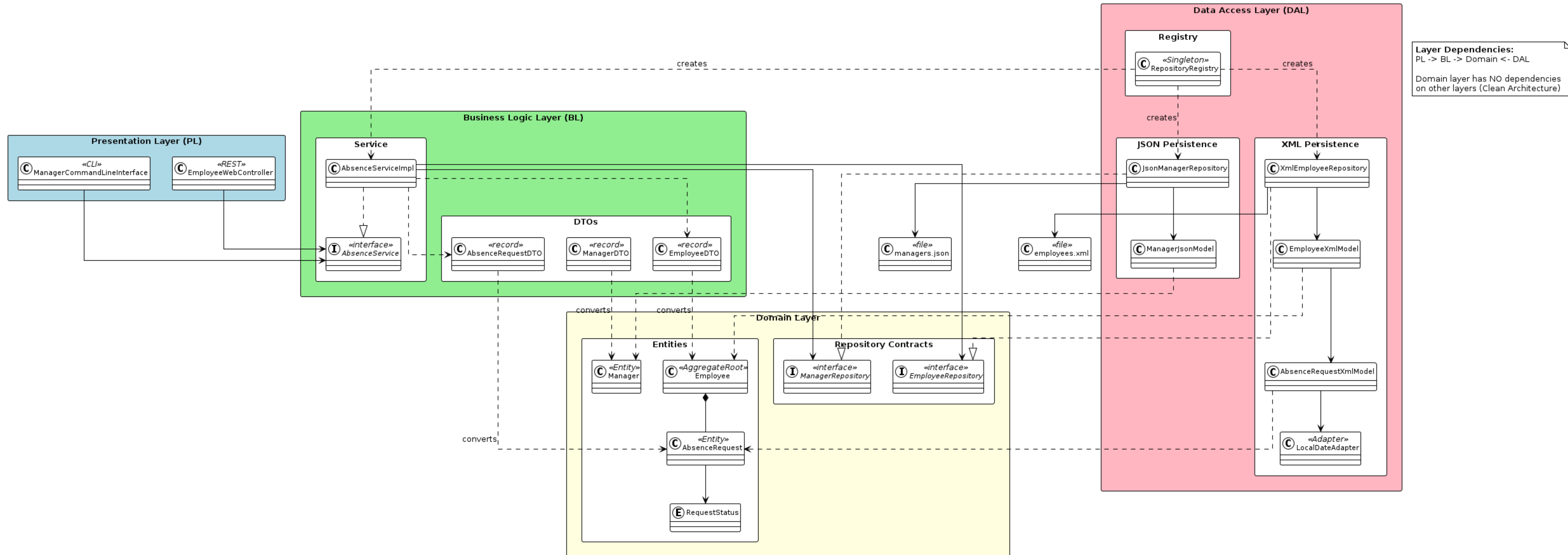
Data Layer - XML/JSON Models (Anti-Corruption Layer)



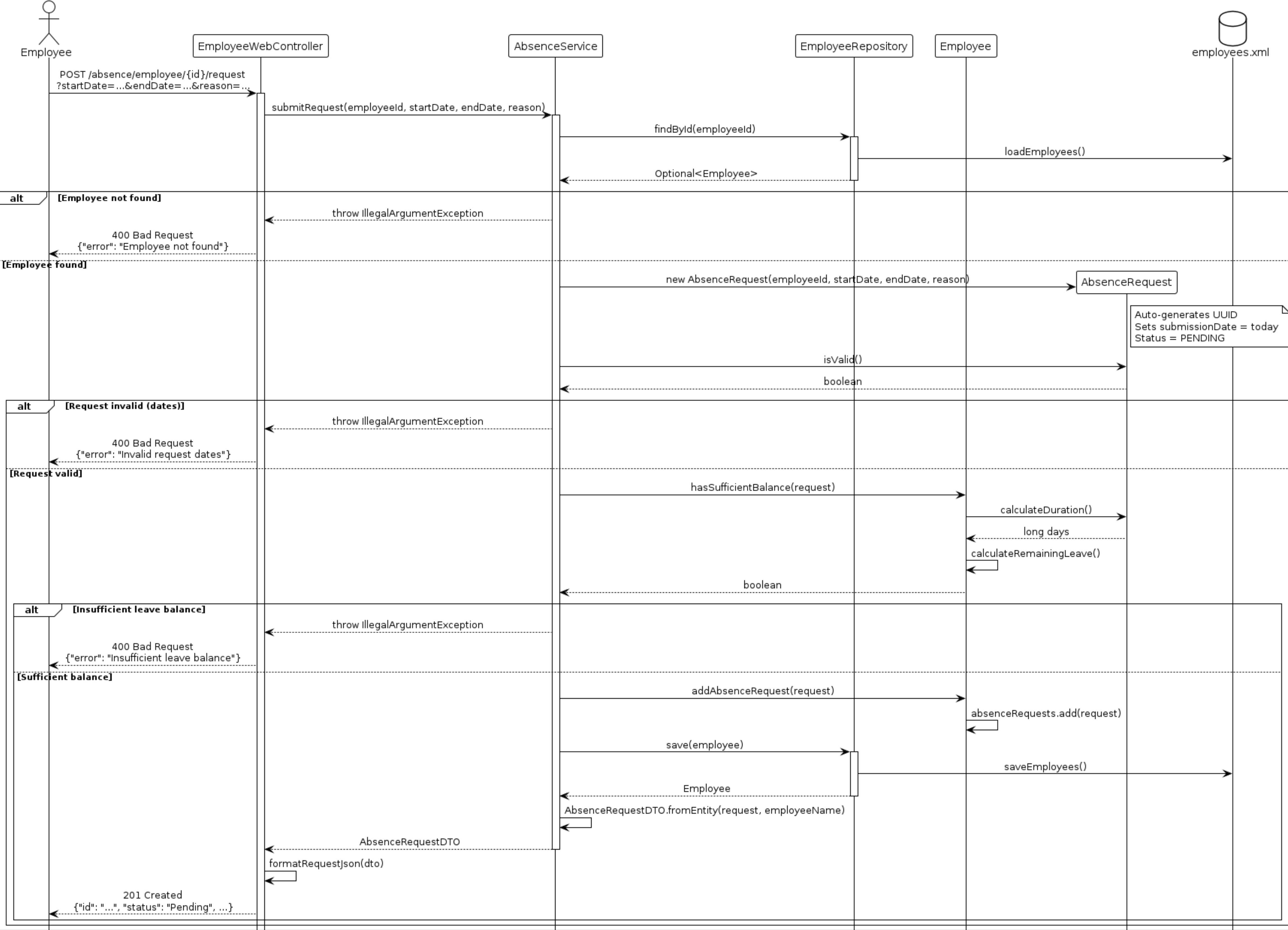
Composition Root - Registry Pattern



Employee Absence Request System - Complete Architecture

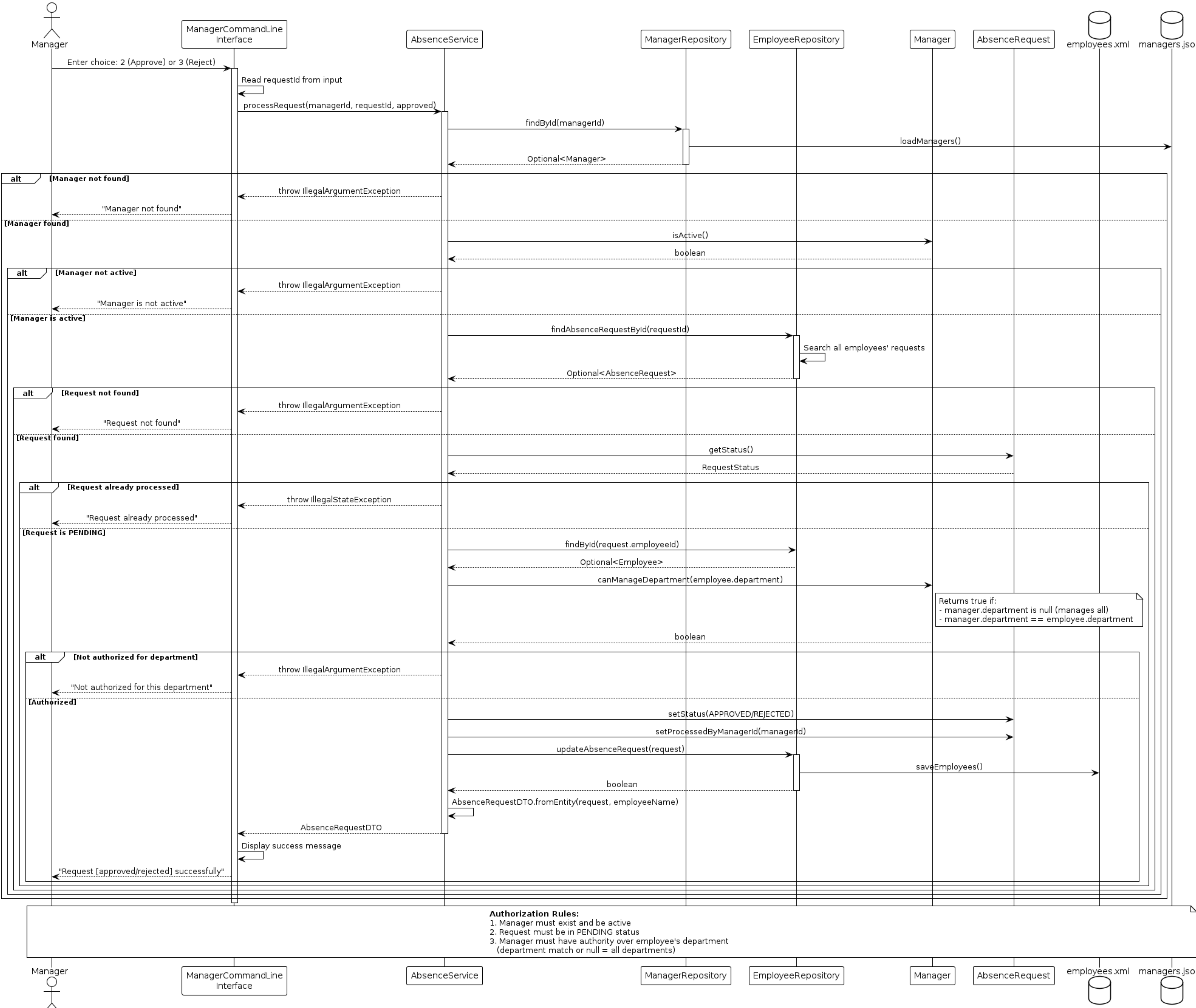


Sequence Diagram: Employee Submits Absence Request (Web)

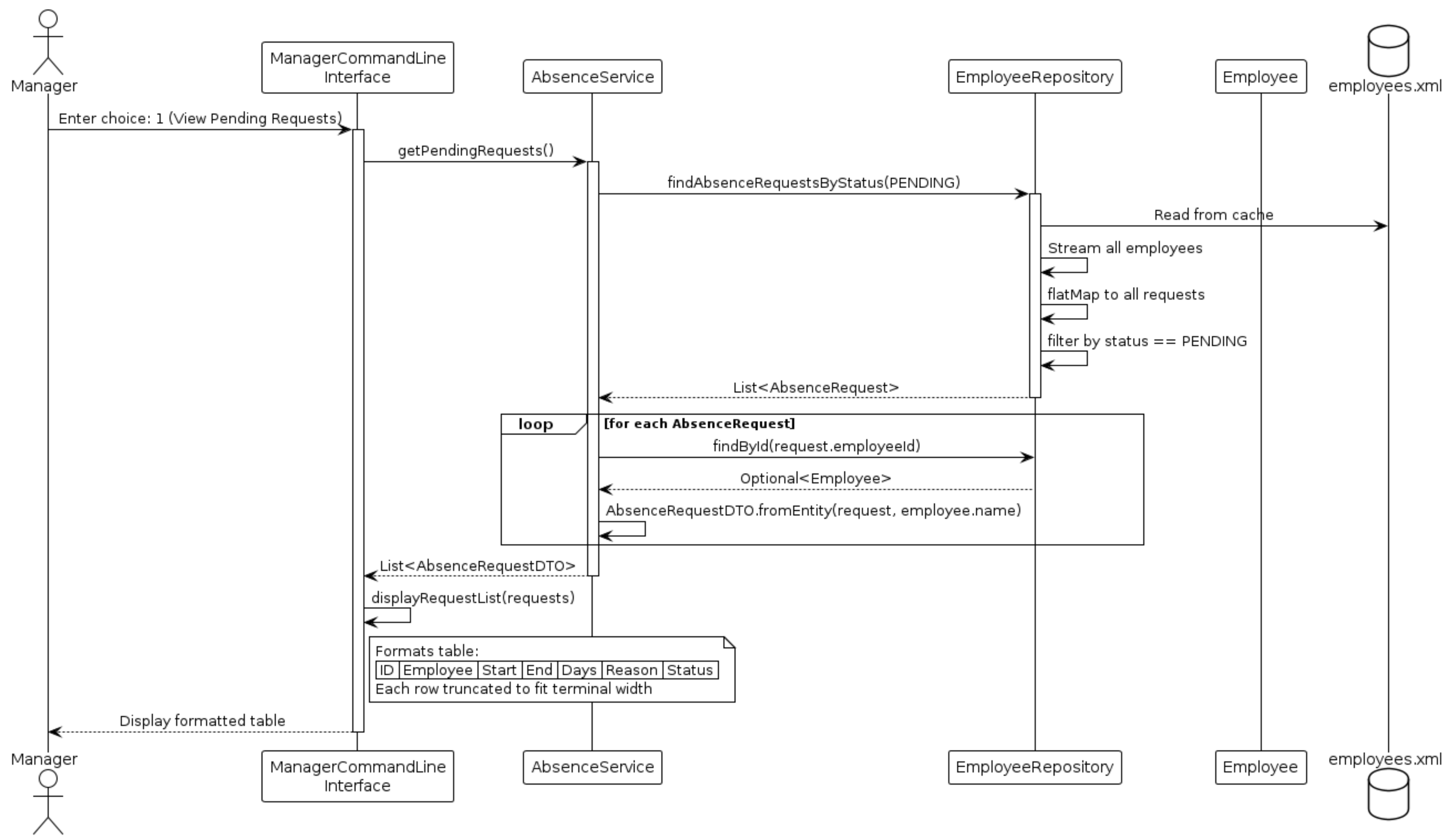


Business Rules Applied:
1. Employee must exist
2. Start date >= today
3. End date >= start date
4. Sufficient leave balance

Sequence Diagram: Manager Processes Absence Request (CLI)



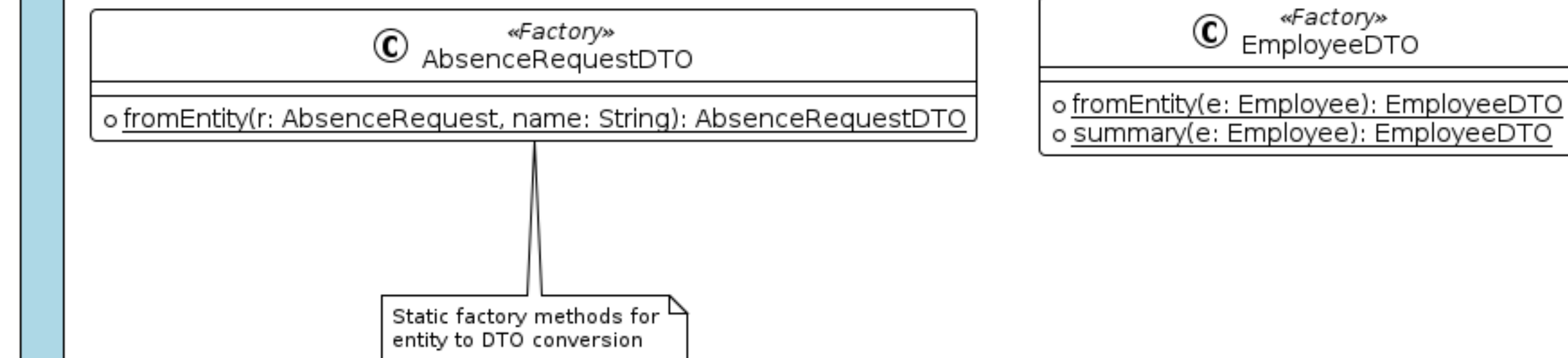
Sequence Diagram: Manager Views Pending Requests (CLI)



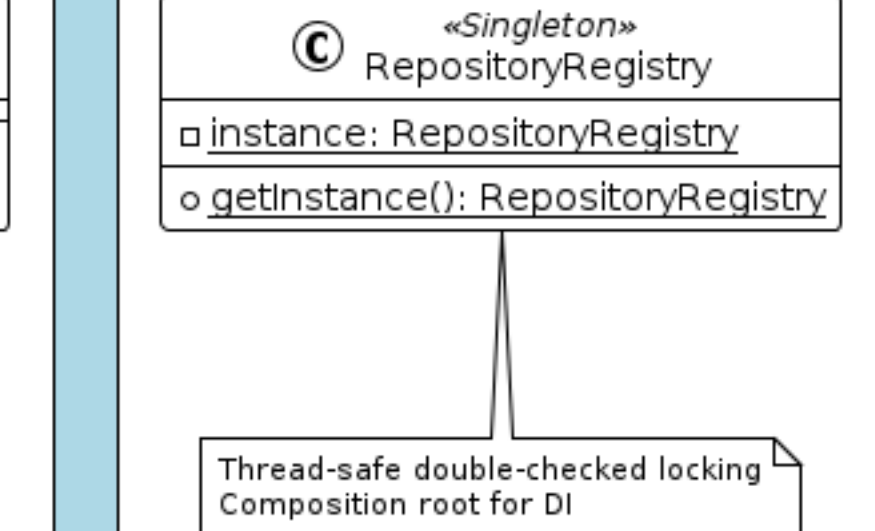
Design Patterns Used in Employee Absence Request System

Creational Patterns

Factory Method Pattern

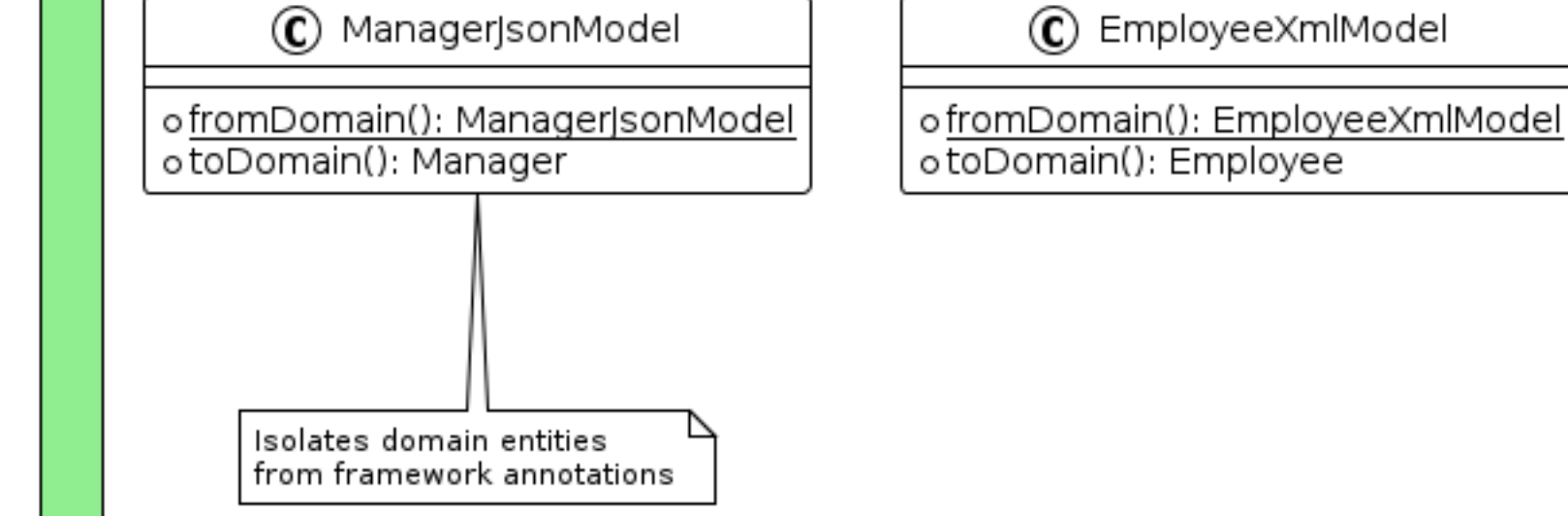


Singleton Pattern

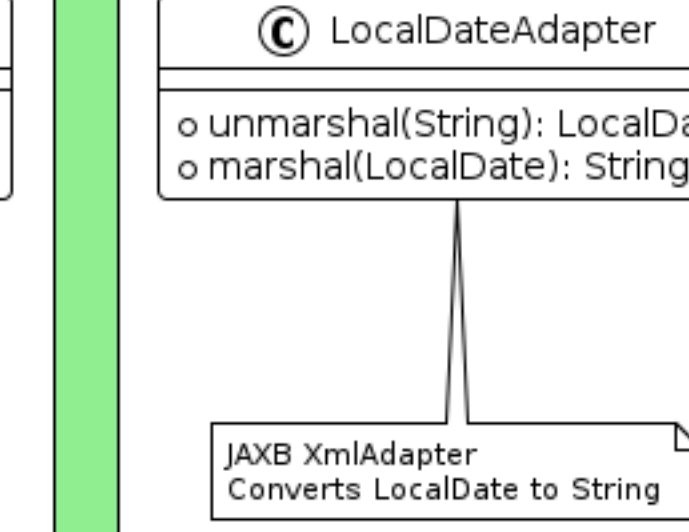


Structural Patterns

Anti-Corruption Layer

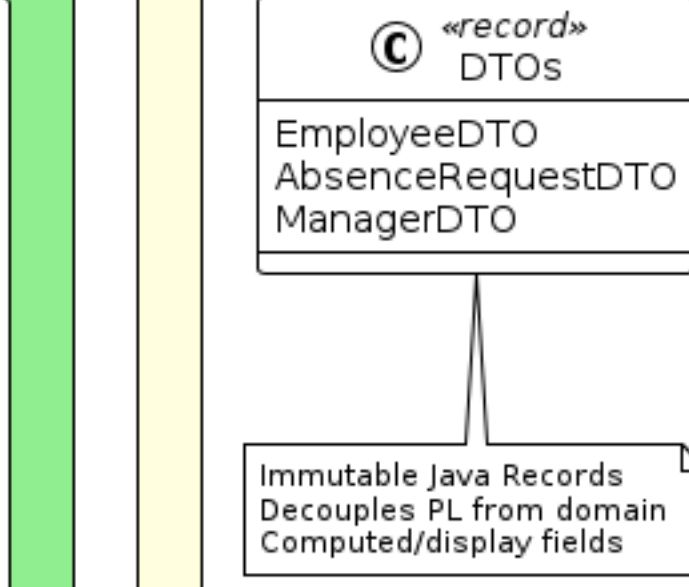


Adapter Pattern

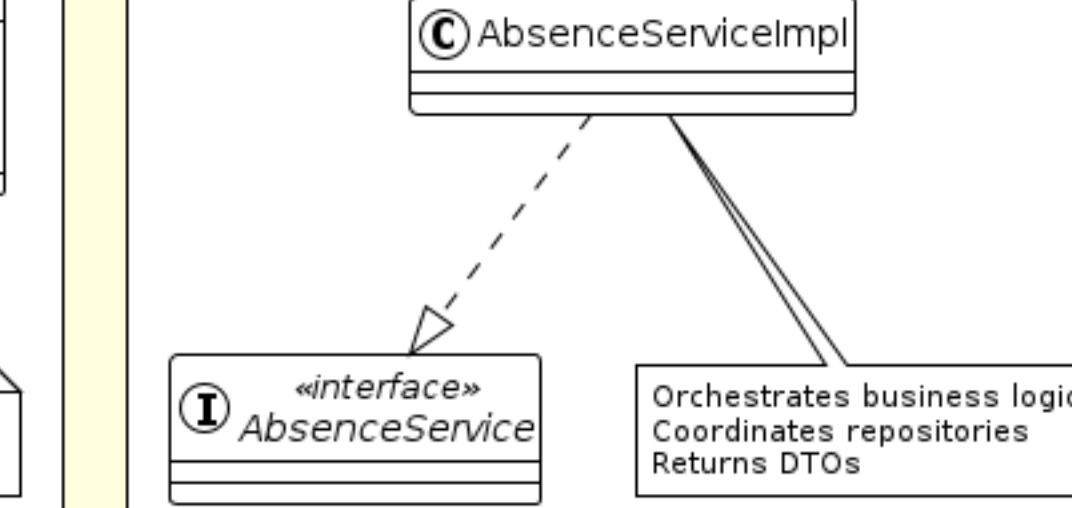


Behavioral Patterns

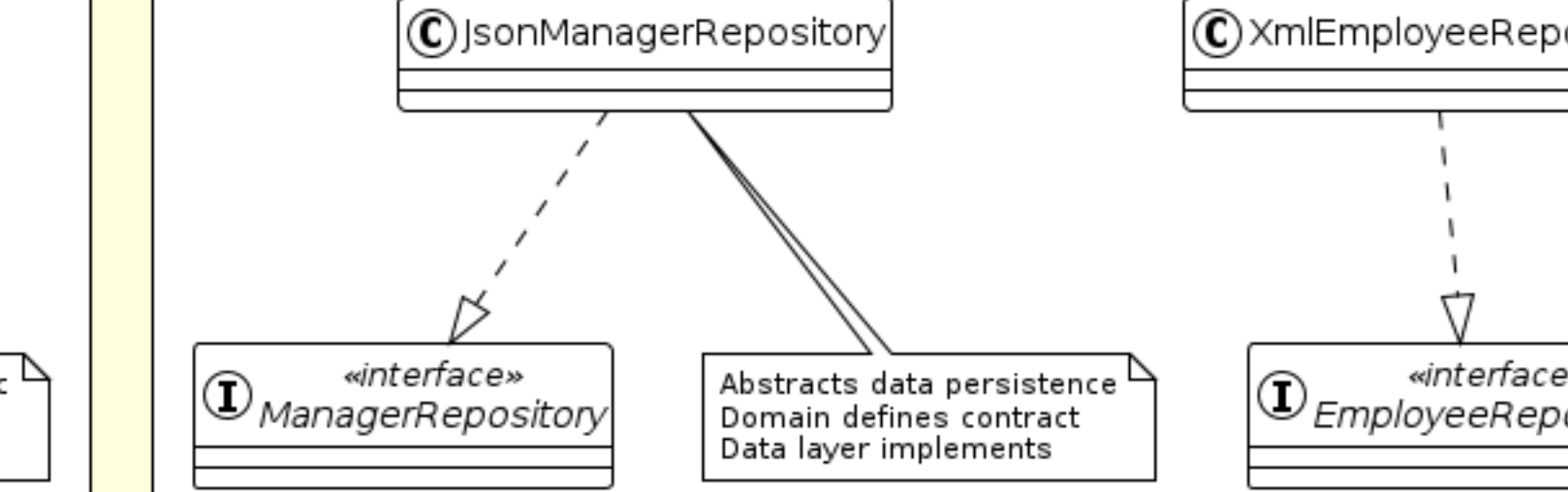
DTO Pattern



Service Layer Pattern

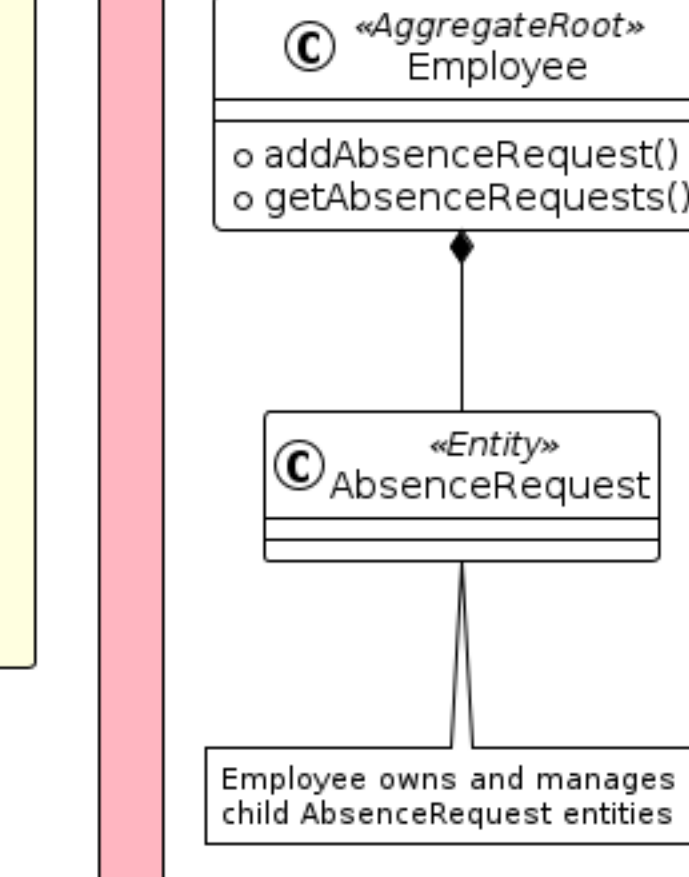


Repository Pattern

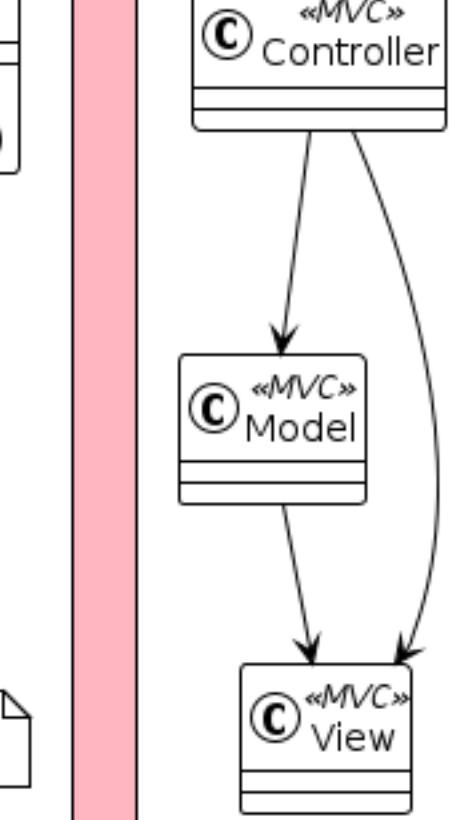


Architectural Patterns

Aggregate Root



MVC Pattern



Three-Tier Architecture

